INCH-POUND

MIL-C-70532 (AR)

29 OCTOBER 1990

MILITARY SPECIFICATION

CARTRIDGES, 105MM, HE, RA, XM913 AND XM927 (WITH WARHEAD AND MOTOR BODY) LOADING, ASSEMBLING AND PACKING

This Specification is approved for use by the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains requirements and quality assurance provisions for the loading, assembling and packing for the cartridges, 105MM, HE, RA, XM913 and XM927 including loading of the motor body and warhead (see note 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-P-46565	- Primer Percussion, M28A2 and M28B2, Loading, Assembling and Packing
MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1320

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STANDARDS

MILITARY

MIL-STD-129	- Marking for Storage and Shipment
MIL-STD-410	- Nondestructive Testing Personnel Qualification and Certification
MIL-STD-453	- Inspection, Radiographic
MIL-STD-1167	- Ammunition Data Card

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from: Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (see 6.9)

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

PRODUCT AND PACKAGING DRAWINGS

9390990	- Cartridge, 105MM, HERA, XM913
9381135	- Seal, Shear
9391033	- Cartridge, 105MM, HERA, XM927
12624495	- Container, Ammunition, Fiber, Assembly, for Cartridge, 105MM Howitzer, HERA, XM913 and XM927
12624496	- Packing and Marking for Container, Fiber, Ammunition, PA111 for Cartridge, 105MM Howitzer, HERA, XM913 or XM927
12630699	- Charge, Propelling 105MM, XM229
12934900	- Packing and Marking For Container Ammo Metal, PA117, For CTG, 105MM, HERA, XM913 or CTG, 105MM, HERA, XM927
12934903	- Container PA117, and Cushion assembly for CTG, 105MM HERA, XM912 or CTG, 105MM, HERA, XM927
19-48-4231/ 45-20PM1006	- Unitization Procedures for Ammunitions Components Packed in a Cyclindrical Metal or Plastic Container on 4 Way Entry Metal Pallets

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(Copies of drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials and parts shall be in accordance with the applicable drawings and specifications.

3.1.1 Clean explosive scrap. Clean explosive scrap produced in regular operation may be remelted and reused provided that it complies with the requirements for the grade of high explosive specified, except for form. Scraps from floor sweepings, ventilation and suction apparatus (other than from drilling operations specifically controlled to prevent contamination of explosives) shall not be used (see 6.5).

3.2 Assembled cartridge. The assembled XM913 cartridge shall comply with all requirements specified on drawing (dwg.) 9390990. The assembled XM927 cartridge shall comply with all requirements specified on dwg. 9391033.

3.2.1 Motor body. No foreign matter shall be within the interior of the motor body prior to loading of the propellant grains.

3.2.2 Delay. There shall be no lubricant compound on the face of the delay assembly or rocket selector cap assembly.

3.2.3 Propellant grains. The surface of the propellant grains shall be free of chips, cracks, gouges, dirt, grease or other foreign material.

3.2.4 Rotating band. The rotating bands of assembled cartridges shall not have nicks or digs in excess of 1/4 the band width. Surface scratches without evidence of raised copper are acceptable. No nicks or digs shall cause the maximum band diameter to be exceeded.

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3.2.3 Propellant grains. The surface of the propellant grains shall be free of chips, cracks, gouges, dirt, grease or other foreign material.

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3.3 Cavitation in high explosive charge.

3.3.1 Cracks and projected cavities. Cracks and cavities within the explosive charge shall not exceed the requirements specified in Table I when determined in accordance with 4.4.3.1. Cavities having a maximum (max) projected length of 1/32 inch (in) or less shall be disregarded.

TABLE I. Cast defects

	SEGMENTS (See Figure 1)			
	A	B	C	D
Sum of projected areas of the cavities, excluding pipes, cracks and annular rings, square inch (sq. in.)	*1/64	1/4	1/2	1/2
Projected length of any cavity, excluding pipes, cracks and annular rings (in.)	1/8	1/2	1/2	3/4
Piping cavities, (max.) projected area (sq. in.)	Ø	1/4	1/2	----
Piping cavities, max. projected width (in.)	Ø	1/4	1/4	----
Cracks, max. projected width (in.)	1/32	1/32	1/16	----
Annular rings, max. projected width (in.)	Ø	Ø	1/4	----

* If the length of the largest cavity is 1/16 inch or less, the maximum total projected area may be 1/20 square inch. Piping cavitation is defined as cavitation located on or near the vertical center line of the projectile with the longitudinal axis (length) three times or greater than the width.

An annular ring is defined as cavitation that takes the form of a ring or portion thereof around the periphery of the cast.

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3.3.2 Surface cavitation. Cavities in Segment D (see Figure 1) formed due to chipped or broken explosive from the sidewalls of the fuze well shall not extend in aggregate around more than 1/4 of the circumference. Cavities in the base of the fuze well shall not be cause for rejection provided the sum of the areas of the cavities is not greater than 80 percent of the area of the fuze well base and no individual point extends more than 1/4 inch below the maximum depth of the fuze well permitted by the drawing, and provided no point extends above the flat surface on which the liner would rest. (Cavities having a max. dimension of 1/32 inch or less shall be disregarded in this computation as shall any voids above Segment D).

3.3.3 Porous areas. Porous areas shall be treated as cavities and shall be subject to the restrictions placed upon cavities except that 80 percent of the projected length and 80 percent of the projected area shall be considered for acceptance purposes.

3.3.4 Cracked charges. Not more than one transverse crack shall be permitted in Segment A and not more than a total of the two transverse cracks shall be permitted in Segment A, B and C combined. Cracks shall be disregarded in Segment D.

3.3.5 Base separation. There shall be no base separation in excess of fifteen thousands of an inch (0.015) between the base of the shell cavity and the cast when determined in accordance with 4.5.3.

3.3.6 Metal parts defects. There shall be no cracks, pipes or visually apparent thin wall sections in the warhead (see 4.4.3.1.2).

3.3.7 Foreign material. There shall be no identifiable foreign material in the explosive charge (4.4.3.1.3).

3.4 Loading of high explosives.

3.4.1 Mating surfaces. The mating surfaces of threads shall be free of caked explosive and all threads shall be free of foreign material. If a chemical cleaner is required, acetone or methylene chloride shall be used. Care shall be taken to prevent the chemical cleaner from coming into contact with the charge and becoming entrapped in the warhead.

3.4.2 Solid explosives. Solid explosives shall not be in the warhead before loading or added to the warhead during initial pouring or cooling.

3.4.3 Coating. Warheads having the interior wall imperfectly coated shall not be loaded. The interior surface of the warhead shall be dry when the high explosive is poured into the warhead (see 4.4.2.1).

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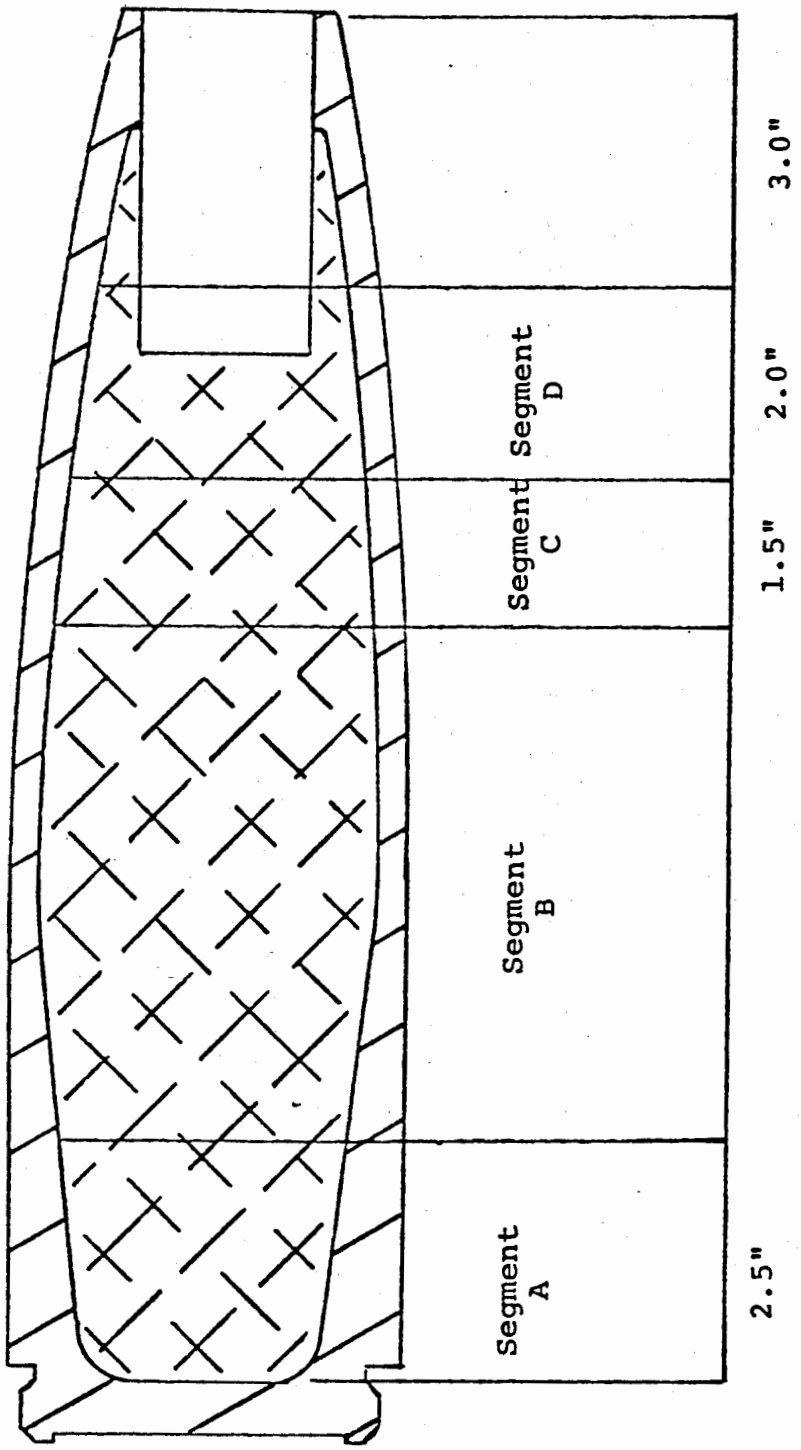


FIGURE 1. Proj. 105MM, HERA, XM913 Warhead Segments.

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3.4.4 Warhead interior cavity. The interior cavity of the warhead prior to loading shall be free from foreign matter.

3.4.5 Fuze well. The fuze well shall be free from any foreign matter or loose explosive prior to assembly of the liner.

3.4.6 Unsolidified pool of paint. There shall be no unsolidified pool of paint in the bottom of the cavity prior to loading of the explosive.

3.4.7 Exterior surface. There shall be no explosive on the exterior surface of the warhead after loading.

3.5 Assembly torque. The motor body shall be assembled to the warhead while the adhesive is still wet to meet the torque requirement specified on dwg. 9381147/12926808. The equipment used for applying torque to the motor body-warhead assembly shall be calibrated at the start of each production shift as well as each time there is a change in equipment, settings or process that is likely to affect uniformity of performance as determined by the Government. The equipment shall be provided with a suitable indicator to aid in observance of proper torque application (see 4.4.3.2). The calibration requirement shall apply to automatic equipment only. Manual torque wrenches shall be calibrated in accordance with standard procedures and intervals for the type of wrench used.

3.6 proving ground test. The projectile shall comply with the following requirements when tested as specified in 4.4.3.4.

3.6.1 Functioning.

3.6.1.1 Rocket motor functioning. When the rocket cap is removed (rocket motor to function), the rocket shall fire in flight. The rocket motor shall ignite not earlier than nine seconds after firing. Lots with early motor ignitions, which pass all other proving ground test requirements, shall be referred to the technical agency for disposition.

3.6.1.2 Precision.

3.6.1.2.1 Pooled range probable error. The range probable error (RPE) of the ballistic samples fired as specified in 4.4.3.4.1 shall not exceed the appropriate value listed below:

Degrees-of-freedom	Maximum RPE (percent)
9	.514
8	.530
7	.550
6	.577
5	.616

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3.6.1.2.2 Horizontal range. No projectile shall have a horizontal range less than 97.5 percent of the mean horizontal range, unless the rocket motor fails to ignite.

3.6.1.2.3 Muzzle velocity. The muzzle velocity for the cartridge, fired at 70° F, shall not be more than 2160 feet per second nor less than 2110 feet per second and the standard deviation of the lot shall not be greater than 8 feet per second (2.44 meters per second).

3.6.1.3 Security. There shall be no premature burst or indication of metal parts separation in the gun bore or in flight.

3.6.2 Safety. There shall be no premature burst or indication of metal parts separation in the gun bore, or in flight, when fired at any of the conditions specified in 4.4.3.4. In addition, no projectile fired in the safety phase shall have a horizontal range less than 97.5 percent of the mean unless the rocket motor fails to ignite.

3.6.2.1 Pressure. The pressure of any individual cartridge, fired at +145° F, shall not produce a PIMP (Permissible Individual Maximum Pressure) greater than 57,000 psi when fired in a M119 Howitzer.

3.7 Ring gaging. A ring gage manufactured to maximum bourrelet diameter shall pass over the warhead and the rocket motor warhead joint after final touch-up painting.

3.8 Radiographic film. Only a film type and vendor as listed herein has been tested and approved for use in radiographing this projectile. A substitute item shall not be used before prior testing and approval. (See 6.9.1)

3.9 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of the first article samples by the contractor shall be as specified in the contract.

3.10 Workmanship. The requirements of workmanship are as shown on the applicable drawing, in the applicable specifications and the following:

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3.10.1 Cleaning. If required, the cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agent.

3.10.2 Protection of the rotating band. Care shall be exercised at all times so that the rotating band of each motor body will not be nicked or burred.

3.10.3 Painting. Painting shall comply with the requirements of the applicable drawing. All paint shall be dry to the touch before testing and packing for shipment. Drying time prior to testing shall be in accordance with the applicable specification.

3.10.4 Foreign matter. No part of assembly shall contain dirt, grease, chips, rust corrosion or other foreign matter, prior to inspection and shipping.

3.10.5 Flash reducer. The propellant and potassium sulfate shall be free of dirt, foreign material and other defects that would make the item unsuitable for its intended use. The cloth and thread used for the manufacture of flash reducer bags shall be void of grease, oil, dirt, foreign material, holes, rips or tears. All components and assemblies shall be fabricated and finished in a thorough workmanlike manner. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contractor or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspection are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all

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requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. The following types of inspection shall be conducted on this item:

a. First article inspection.
b. Quality conformance inspection.

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated:

Part description	Drawing	Quantity
Cartridge	9390990/9391033	10
Projectile	9381147	10
Warhead Loading Assembly	9381152	10
Shear, Seal	9381135	10
Flash reducer	12630698	10

4.3.2 Inspection to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I. First article inspection

CLASSIFICATION OF CHARACTERISTICS

MIL-C-70532 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
	Cartridges, 105MM, HERA, XM913/XM927		See below
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Cartridges, 105MM, HE, RA, XM913/XM927			
	Loading, Assembly and Packing			
	(Dwg. 9390990/9391033)			
	Examination for defects	10	3.2	4.4.2.16
	Examination for defects	a	3.2	4.4.2.2 thru 4.4.2.15
	Seal Shear			
	(Dwg. 9381135)			
	Examination for defects	10	3.2	4.4.2.17
	Flash Reducer			
	(Dwg. 12630698)			
	Examination for defects	10	3.2	4.4.2.18

NOTES: a. Two of the cartridges submitted for first article shall be disassembled to allow checking of the examinations from 4.4.2.1 through 4.4.2.15.

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TABLE I. First article inspection

CLASSIFICATION OF CHARACTERISTICS

MIL-C-70532 (AR)

PARAGRAPH	TITLE Packaging for Cartridges, 105MM, HERA XM913/XM927	SHEET 2 OF 2	DRAWING NUMBER See below
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Packaging and Marking for Container, Fiber, Ammo, PA111 for Cartridge, 105MM, How, HERA, XM913 or XM927 (Dwg. 12624496)			
	Examination for defects	10	3.1	4.4.2.21/4.4.2.22
	Packaging and Marking for Container Ammo Metal, PA117, for Cartridge, 105MM, HERA, XM913 or 105MM HERA XM927 (Dwg. 12934900)			
	Examination for defects	10	3.1	4.4.2.23/4.4.2.24

NOTES:

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall conform with the formation provisions of MIL-A-48078. In addition, each inspection lot of loaded cartridges shall contain:

a. Warhead metal parts of one lot interfix number from one manufacturer.

b. TNT of the same type from one lot interfix number from one manufacturer.

c. Supplementary charges of one lot interfix number from one manufacturer.

d. Cartridge cases of one lot interfix number from one manufacturer.

e. Motor body metal parts of one lot interfix number from one manufacturer.

f. Rocket propellant grains from up to four (4) consecutively produced lots of one lot interfix number from one manufacturer.

g. Primer from one lot interfix number from one manufacturer.

h. Delay assemblies of one lot from one manufacturer.

i. Propelling charges from not more than one lot.

j. Shear seal from the same drawing, drawing revision and manufacturer.

k. Potassium sulfate from not more than one lot.

l. M30 propellant from not more than one lot.

4.4.2 Examinations and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified herein. Attributes Sampling Inspection shall be conducted in accordance with TABLE II below, using the inspection levels cited in the Classification of Characteristics paragraphs:

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TABLE II. Ø-1 Attributes Plans

Lot Size	I	II	III	IV	V	VI
2 to 8	*	*	*	*	5	3
9 to 15	*	*	*	13	5	3
15 to 25	*	*	*	13	5	3
26 to 5Ø	*	*	32	13	5	3
51 to 9Ø	*	*	32	13	12	4
91 to 15Ø	*	125	32	13	12	5
151 to 28Ø	*	125	32	3Ø	14	6
281 to 5ØØ	*	125	32	3Ø	17	7
5Ø1 to 12ØØ	*	125	74	35	2Ø	9
12Ø1 to 32ØØ	125Ø	125	74	43	24	1Ø
32Ø1 to 1ØØØØ	125Ø	125	87	5Ø	3Ø	1Ø
1ØØØ1 to 35ØØØ	125Ø	296	1Ø9	61	36	1Ø

Numbers under inspections levels indicate sample size; asterisks indicate one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

b. Alternative inspection provisions. Alternative inspection procedures, methods or equipment such as statistical process control, tool control, other types of sampling procedures etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the procuring contracting officer for evaluation and approval by the Government. When required, the contractor shall demonstrate that the effectiveness of the proposed alternative(s) is equal to or better than the specified quality assurance provisions herein. In cases of dispute as to whether the contractor's proposed alternative(s) provide equal assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into contractor's quality program or detailed inspection system, as applicable.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-70532 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Warhead Loading Assembly Prior to Loading	1 OF 1	9381152
			NEXT HIGHER ASSEMBLY 9381147

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Special				
1	Foreign matter in charge cavity	100%	3.4.4	Visual
2	Unsolidified pool of paint or blisters in warhead cavity	100%	3.4.6	Visual/Manual
Major				
101	Interior protective coating with bare spots	Level III	3.4.3	Visual
Minor				
201	Evidence of poor workmanship	Level V	3.10	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-70532 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Warhead Loading Assembly After Loading	1 OF 1	9381152
			NEXT HIGHER ASSEMBLY 9381147

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Special				
1	Presence of explosive on exterior of shell body including nose, threads	100%	3.4.7	Visual
Major				
101	Depth of fuze well	Level III	3.2	Gage
102	Foreign matter or loose explosive in fuze well	Level III	3.4.5	Visual
Minor				
201	Diameter of fuze well	Level V	3.2	Gage
202	Surface cavitation excessive	Level V	3.3.2	Visual
203	Evidence of poor workmanship	Level V	3.10	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-70532 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Warhead Loading Assembly Prior to Assembly of Supplementary Charge	1 OF 1	9381152
			NEXT HIGHER ASSEMBLY: 9381147

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	True position of inside diameter of liner with minor diameter of internal thread and end face of assembly	Level III	3.2	Gage
102	Depth to bottom of liner	Level III	3.2	Gage
103	Swage missing or incomplete	100%	3.2	Visual
104	Inside diameter of liner after swaging	Level III	3.2	Gage
Minor				
201	Liner damaged	Level V	3.2	Visual
202	Foreign matter in liner cavity including explosive	Level V	3.2	Visual
203	Evidence of poor workmanship	Level V	3.10	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-70532 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Warhead Loading Assembly, Prior to Assembly of Shipping Plug	1 OF 1	9381152
			NEXT HIGHER ASSEMBLY 9381147

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Supplementary charge pad or supplementary charge missing	100%	3.2	Gage
Major				
101	Supplementary charge inverted	Level III	3.2	Visual
Minor				
201	Supplementary charge damaged	Level V	3.2	Visual
202	Foreign matter on threads including explosive	Level V	3.4.1	Visual
203	Spacer missing or damaged if applicable	Level V	3.2	Visual
204	Corrosion preventative compound missing from threads	Level V	3.2	Visual
205	Evidence of poor workmanship	Level V	3.10	Visual

NOTES:

AMSMC Form 1570b, 1 Jul 89

Replaces 1570, 1 Feb 85, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-70532 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Delay Assembly, Prior to Assembly to Motorbody	1 OF 1	9381147
			NEXT HIGHER ASSEMBLY
			9391033/9390990

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	O-ring missing	Level III	3.2	Visual
102	Lubricant missing or inadequate on O-ring	Level III	3.2	Visual
Minor				
201	Evidence of poor workmanship	Level V	3.10	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-70532 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Motor Body, Prior to Assembly of Delay Assembly	1 OF 1	9381147
			NEXT HIGHER ASSEMBLY: 9390990/9391033

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Special 1	Nozzle insert missing, loose or damaged	100%	3.2	Visual/Manual
Major	None defined			
Minor	None defined			

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-70532 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Delay Assembly, After Assembly to Motor Body	1 OF 1	9381147
			NEXT HIGHER ASSEMBLY: 9390990/9391033

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Shear seal missing or not pressed	Level III	3.2	Visual
102	Lubricant on face of delay assembly	Level III	3.2.2	Visual
Minor				
201	Evidence of poor workmanship	Level V	3.10	Visual

NOTES:

AMSMC Form 1570b, 1 Jul 89 — Replaces 1570, 1 Feb 85, which may not be used.

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CLASSIFICATION OF CHARACTERISTICS

MIL-C-70532 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Rocket Selector Cap Assembly, Prior to Assembly to Motor Body	1 OF 1	9381147
			NEXT HIGHER ASSEMBLY 9390990/9391033

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Rocket selector cap O-ring or gasket missing	100%	3.2	Visual
102	Lubricant not applied to O-ring	100%	3.2	Visual
103	Lubricant on face of rocket selector cap assembly	Level III	3.2.2	Visual
Minor				
201	Evidence of poor workmanship	Level V	3.10	Visual

NOTES:

AMSMC Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used.

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CLASSIFICATION OF CHARACTERISTICS

MIL-C-70532 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.9	Motor Body, Prior to Loading Propellant Grain	1 OF 1	9381147
			NEXT HIGHER ASSEMBLY: 9390990/9391033

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Foreign matter in cavity	Level III	3.2.1	Visual
102	Grain spacers missing or obviously improperly assembled	Level III	3.2	Visual
103	Presence of any chips, cracks, gouges, dirt, grease or other foreign material on surface of the grain	Level III	3.2.2	Visual
Minor				
201	Evidence of poor workmanship	Level V	3.10 Excl. 3.10.4	Visual
. . .				

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CLASSIFICATION OF CHARACTERISTICS

MIL-C-70532 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.10	Motor Body and Warhead, Prior to Assembly	1 OF 1	9381147
			NEXT HIGHER ASSEMBLY: 9390990/9391033

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Special				
1	Knurl & threads not free from grease, explosive or foreign material, prior to application of sealant	100%	3.2	Visual
Major				
101	O-ring missing from warhead	Level III	3.2	Visual
102	Warhead insulation missing	Level III	3.2	Visual
103	Surface primer or sealing compound missing or improperly applied to motor body/warhead threads	Level III	3.2	Visual
104	Lubricant not applied to O-ring	Level III	3.2	Visual
Minor				
201	Evidence of poor workmanship	Level V	3.10 Excl. 3.10.4	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-70532 (AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.11	Projectile Assembly	1	1	9381147

NEXT HIGHER ASSEMBLY: 9390990/9391033

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Excessive gap between warhead and motor body assembly	100%	3.2	Gage
102	Projectile fails to pass through ring gage	4.4.3.5	3.7	Gage
103	Rotating band nicked or damaged	Level III	3.2.4	Visual
104	Shipping plug missing or not fully seated	Level III	3.2	Visual
105	Shipping plug washer missing, if applicable	Level III	3.2	Visual
106	Torque test failure of shipping plug, if applicable (loosening at min. disasssembly torque)	Level III	3.2	Test/4.5.2
107	Torque test failure of rocket selector cap	Level III	3.2	Test/4.4.3.6/4.5.2
108	Torque test failure of shipping plug, if applicable (failure to loosen at max. disassembly torque)	Level III	3.2	Test/4.5.2
109	Rocket motor and warhead assembly torque	4.4.3.2	3.5	Visual
110	Re-application of assembly torque to rocket motor/warhead joint	4.4.3.2	3.5	Test/4.5.2
Minor				
201	Evidence of poor workmanship	Level V	3.10	Visual

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CLASSIFICATION OF CHARACTERISTICS

MIL-C-70532 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.12	Primer, Prior to Assembly to Case	1 OF 1	9939130
			NEXT HIGHER ASSEMBLY
			9390990/9391033

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical 1	One or more flash holes missing	100%	3.2	Gage
Major 101	Check test of primer	4.4.3.5	3.2	Test
Minor	None defined			

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CLASSIFICATION OF CHARACTERISTICS

MIL-C-70532 (AR

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.13	Cartridge Case, Prior to Assembling Propelling Charge	1 OF 1	9390990/9391033
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Primer above flush	100%	3.2	Gage
Major				
101	Primer missing or damaged	Level III	3.2	Visual
Minor				
201	Foreign matter in interior of case	Level V	3.2	Visual
202	Evidence of poor workmanship	Level V	3.10	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-C-70532 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.14	Propelling Charge, Prior to Assembly into Cartridge Case	1 OF 1	9205472/9382042
			NEXT HIGHER ASSEMBLY 9390990/9391033

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Increment missing (M67 charge)	100%	3.2	Gage
Major				
101	Any opening in bag large enough to permit escape of propellant	Level III	3.2	Visual
Minor				
201	Cord broken (M67 charge)	Level V	3.2	Visual
202	Evidence of poor workmanship	Level V	3.10	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.15	Cartridge Case With Propelling Charge	1 OF 1	9391033
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Flash reducer missing or insecure (XM229)	100%	3.2	Manual
Major				
101	Incorrect sequence or increments (M67)	100%	3.2	Visual
102	Foil side of bag, charge 5, not facing primer (M67)	Level III	3.2	Visual
Minor	None defined			

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.16	Projectile and Cartridge Case Assemblies, Prior to Pack	1 OF 1	9390990/9391033
			NEXT HIGHER ASSEMBLY 12624496

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major 101	Any component missing	Level III	3.2	Visual
Minor 201	Exterior coating of shell with an aggregate of bare spots in excess of 1/4 square inch	Level V	3.2	Visual
202	Marking misleading, unidentifiable or missing including incorrect color	Level V	3.2	Visual
203	Evidence of poor workmanship	Level V	3.10	Visual

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CLASSIFICATION OF CHARACTERISTICS

MIL-C-70532 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.17	Seal, Shear	1 OF 1	9381135
			NEXT HIGHER ASSEMBLY
			9390990/9391033

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Length	Level III	3.2	Gage
102	True position of inside diameter to outside diameter	Level III	3.2	Gage
103	Outside diameter	Level III	3.2	Gage
Minor				
201	Evidence of poor workmanship	Level V	3.10	Visual

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CLASSIFICATION OF CHARACTERISTICS

MIL-C-70532 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.18	Body, Flash Reducer	1 OF 1	12630696
			NEXT HIGHER ASSEMBLY 12630697

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Cloth improper	Level III	3.1	4.5.5
102	Diameter of bag	Level III	3.2	Gage/Scale
Minor				
201	Marking missing, misleading or unidentifiable	Level V	3.2	Visual
202	Evidence of poor workmanship	Level V	3.10	Visual

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CLASSIFICATION OF CHARACTERISTICS

MIL-C-70532 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.19	Flash Reducer	1 OF 1	12630698
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Weight of M30 propellant	100%	3.1	Scale/4.5.6
102	Weight of potassium sulfate	100%	3.1	Scale/4.5.6
103	Weight of flash reducer	100%	3.1	Scale/4.5.6.1
104	Assembly damaged to the extent that contents can escape	Level III	3.2	Visual
105	Any seam or opening incompletely stitched	Level III	3.2	Visual
106	Thread broken	Level III	3.2	Visual
Minor				
201	Evidence of poor workmanship	Level V	3.10	Visual

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CLASSIFICATION OF CHARACTERISTICS

MIL-P-70532 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.20	Charge, Propelling 105MM XM229	1 OF 1	12630699
			NEXT HIGHER ASSEMBLY 9390990

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Seams incompletely stitched	Level III	3.2	Visual
102	Flash reducer not secure	Level III		Manual
Minor				
201	Marking missing, misleading, or unidentifiable	Level V	3.2	Visual
202	Evidence of poor workmanship	Level V	3.10	Visual

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CLASSIFICATION OF CHARACTERISTICS

MIL-P-70532 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.21	Packaging and Marking for Container, Fiber (unsealed)	1 OF 1	12630699
			NEXT HIGHER ASSEMBLY 12934900

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Packing components missing, damaged, improperly assembled	Level III	3.1	Visual
102	Cartridge cannot be inserted or removed easily by hand	Level III	3.1	Manual
103	Container covers cannot be removed easily by hand	Level III	3.1	Manual
104	Insufficient fillers for tight pack (slight gap between body and covers)	Level III	3.1	Visual
Minor				
201	Evidence of poor workmanship	Level V	3.10	Visual

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CLASSIFICATION OF CHARACTERISTICS

MIL-C-70532 (AR)

PARAGRAPH	TITLE	SHEET OF	DRAWING NUMBER
4.4.2.22	Packing and Marking for Container, Fiber (sealed)	1 1	12624496
			NEXT HIGHER ASSEMBLY 12934900

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Cover seal tape incomplete, wrinkled, or the wrong material	Level III	3.1	Visual
102	Cloth strap incorrect width or length	Level III	3.1	Gage
103	Cloth strap assembled incorrectly	Level III	3.1	Visual
104	Marking misleading, missing, or incorrect	Level III	3.1	Visual
Minor				
201	Tear tape, tab length, min.	Level V	3.1	Gage
202	Evidence of poor workmanship	Level V	3.10	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-P-70532 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.23	Packing and Marking for Container, Fiber (sealed)	1 OF 1	12934900
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Any component missing or damaged	Level III	3.1	Visual
102	Moisture content of material	Level III	3.2	Certification
103	Handle on fiber container missing	Level III	3.1	Visual
104	Insufficient fillers for tight pack	Level III	3.1	Visual/Manual
105	Seal tape on fiber containr peeling. Damaged during insertion/extraction from can	Level III	3.1	Visual/Manual
Minor				
201	Evidence of poor workmanship	Level V	3.10	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-C-70532 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.24	Sealed Metal Container	1 OF 1	12934900
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Container damaged, exposing contents	level III	3.1	Visual
102	Metallic seal missing or improperly secured	Level III	3.1	Visual
103	DOD symbol missing or incorrect	Level III	3.1	Visual
104	Air leak test	100%	3.1	Gage
105	Screw not properly replaced and torqued after test	Level III	3.1	Visual
Minor				
201	Marking incorrect, missing or unidentifiable	Level V	3.2	Visual
202	Contents loose	Level V	3.2	Manual

NOTES:

AMSMC Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used.

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CLASSIFICATION OF CHARACTERISTICS

MIL-C-70532 (AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.25	Unitization Procedure for Cylindrical Metal or Plastic Containrs on 4-way Entry Pallets	1	1	19-48-4231/45-20PM1006
				NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Missing or loose strapping	Level III	3.1	Visual/Manual
102	Incorrect number of containers on pallet (except for light load)	Level III	3.1	Visual
103	Containers not all positioned in the same direction and properly interlocked	Level III	3.1	Visual
104	Marking missing, incorrect or illegible	Level III	3.1	Visual
Minor				
201	Evidence of poor workmanship	Level V	3.10	Visual

NOTES:

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection tables and the following:

4.4.3.1 Cavitation. (See 3.3 except 3.3.2). Cavitation defects are classified as follows:

Critical:

1. Cavitation, porosity, piping, cracks or annular rings in Segment A or B in excess of that permitted in Table I.

2. Cavitation, cracks or annular rings in Segment C in excess of that permitted in Table I.

3. Cracks in excess of that permitted by 3.3.4.

4. Base separation (see 4.4.3.1.1).

5. Cavitation in Segment D in excess of that permitted in Table I.

This test shall be performed on all loaded warhead assemblies (100 percent inspection). The radiographic test shall be performed as specified 4.5.1, using equipment in accordance with 4.4.4. Any assembly which has any one of the above critical defects shall be classed defective and removed from the lot.

Warheads rejected by radiographic examination may be drilled out to one half the depth of the "B" segment in Figure 1 and repoured, but not more than one drilling operation may be performed per warhead.

4.4.3.1.1 Base separation. Warheads showing the largest base separation as determined by Radiographic examination prior to post cycle heat treatment shall be selected for sectioning at the following rate. Select warheads at the rate of one (1) out of seven hundred (700) warheads to be included in the post cycle heat treatment lot with a minimum of two (2) samples per lot. These warheads are to be included in the post cycle heat treatment and section for determination of base separation thereafter. If insufficient samples are available with base separation based on radiographs, a sufficient number of samples shall be randomly selected from the post cycle heat treatment lot to make up the required sample. The samples selected for sectioning shall be placed in the coolest area of the post cycle igloo. Any warhead after sectioning which has base separation in excess of that permitted by the applicable requirement, shall be classed defective, and the entire post cycle heat treatment lot rejected. The test shall be performed as specified in 4.5.3 using equipment in accordance with 4.4.4. Any lot which fails after being post cycle heat treated three times shall be rejected.

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4.4.3.1.2 Metal parts defects. (Critical defect) All radiographs of the warhead section loading assemblies shall be examined for evidence of cracks, pipes and visually apparent thin wall sections in the warhead. The presence of any such indication shall cause that assembly to be classed defective and removed from the lot (see section 6.10).

4.4.3.1.3 Foreign material. (Critical defect) All radiographs or warhead section loading assemblies shall be examined for foreign material in the explosive charge. The presence of any identifiable foreign material shall cause that assembly to classed defective and removed from the lot.

4.4.3.2 Motor body - warhead assembly torque. The torque applied to the motor body-warhead joint shall be witnessed for each projectile during assembly to assure compliance with the applicable requirement. Any projectile improperly assembled shall be classed defective and removed from the lot. In addition, a minimum of one sample projectile shall be randomly selected from each hours production for a check torque application. If any projectile fails to comply with the torque requirements, as evidenced by relative motion at the joint, when tested in accordance with 4.5.2 then all projectiles assembled since the last acceptable sample shall be rejected.

As an alternate to the once an hour torque check sample, a verification check may be made on the torque assembly machine. In this case, the torque readout from the assembly machine shall be verified for transducer accuracy at the beginning and end of the operating shift and after each four-hour production interval. If the torque assembly machine fails the verification test, all the projectiles assembled during that production interval shall be rejected. In the event of two consecutive gap rejects, the torque assembly machine display shall be checked during operation on a subsequent assembly. If the indicated torque is in or below the lower 20% of the acceptance range, a verification check will be made. If the machine fails the verification, all projectiles assembled since the last acceptable verification shall be rejected.

4.4.3.3 Radiographic examination for missing or improperly assembled components. (See dwg. 9381147). This test shall be performed 100 percent. Each assembly shall be subjected to a radiographic examination in a plane that will clearly define all of the Motor Body components, including both channels of the delay assembly. Necessary controls will be implemented to assure that the assembly will be oriented such that both channels are visible when radiographed. The test shall be performed as specified in 4.5.1.2 with equipment in accordance with 4.4.4. Any assembly which shows evidence of missing or improperly assembled components, as listed below, shall be classed as defective and removed from the lot:

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a. Delay housing assembly not properly seated in motor body.

b. Shear ring seal not fully expanded in delay and motor body grooves. The groove in the motor body shall be completely filled with lead. There shall be no gap between the shear ring seal and the shoulder of the expansion cone.

c. Missing Boron Potassium Nitrate charge or Delay Column.

d. Gap between the propellant grain and the motor body base exceeding 0.025 inches maximum as compared to the radiographic standard.

e. Evidence of foreign material present.

f. Any other deviation from the normal assembly condition.

4.4.3.4 Proving ground (see 3.6). These tests shall be performed as specified in 4.5.3. Defects shall be classified as follows:

Categories	Defects
Critical	
1	Security (see 3.6.1.3)
2	Pressure in excess of 57,000 PSI (see 3.6.2.1)
Special	
a	Range less than 97.5 percent of mean horizontal range unless rocket motor fails to ignite (see 6.15 and 3.6.1.2.2)
Major	
101	Range probable error exceeds allowable limit (see 3.6.1.2.1)
102	Rocket motor fails to ignite
103	Muzzle velocity (see 3.6.1.2.3)
Minor	None defined

4.4.3.4.1 Functioning (see 3.6.1). The functioning test shall be conducted in accordance with 4.5.8.1.

4.4.3.4.1.1 Sample size. For first article or beginning with the first lot produced and continuing until three consecutive lots have complied with all the acceptance criteria specified herein, and in 4.4.3.4.2.1, 30 sample projectiles shall be selected at random from each lot for this test and fired with the rocket cap removed (rocket motor to function). If any critical or major defect occurs, the lot shall be rejected.

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4.4.3.4.1.2 Successive lots. After three consecutive lots have complied with 4.4.3.4.1.1, 10 sample projectiles shall be selected at random from each successive lot for this test, and fired with the rocket cap removed (rocket motor to function). If any critical or major defect occurs, the lot shall be rejected. In the event that two successive lots fails to meet any or all of the requirements of this paragraph or 4.4.3.4.2.2, the provisions of 4.4.3.4.1.1 and 4.4.3.4.2.1 shall again apply until three successive lots meet the requirements specified therein.

4.4.3.4.2 Safety (see 3.6.2). The safety tests shall be conducted in accordance with 4.5.3.2.

4.4.3.4.2.1 Sample size. Beginning with the first lot produced and continuing until three consecutive lots have complied with all the acceptance criteria contained herein and in 4.4.3.4.1.1, 30 sample projectiles shall be selected at random from each lot for these tests and fired with the rocket cap removed (rocket motor to function). If any critical defect occurs, the lot shall be rejected.

4.4.3.4.2.2 Successive lots. After three consecutive lots have complied with 4.4.3.4.2.1, 14 sample projectiles shall be selected at random for these tests, and fired with the rocket cap removed (rocket motor to function). If any critical defect occurs, the lot shall be rejected. In the event that two successive lots fail to meet the requirements of this paragraph or 4.4.3.4.1.2, provisions of 4.4.3.4.1.1 and 4.4.3.4.2.1 shall again apply until three successive lots meet the requirements specified therein.

4.4.3.4.2.3 Temperature test failures. Failure of a rocket motor to ignite during +145 degrees F safety tests shall be considered a Major defect, and the lot shall be rejected. If a motor ignition failure occurs during -50 degrees F safety tests, the lot shall be referred to the procuring agency for disposition (see 6.13).

4.4.3.4.3 Test validity. If for any reason the proving ground considers that the test conditions have detrimentally affected the test results, the test shall be considered invalid and a new test shall be performed with additional samples.

4.4.3.5 Ring gage inspection. (Major defect) This test shall be performed as specified in 4.5.4 in accordance with the following plan. At the start of each shift or joint painting operator change this inspection shall be performed on a one hundred (100) percent basis until thirty-two (32) consecutive units have been inspected

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without failure. After this criteria has been met only those projectiles that required touch-up painting of the bourrelet shall be one hundred (100) percent ring gaged. However, in the event of a failure due to excessive paint on the rocket motor-warhead joint, when ring gaging for touched up bourrelets, those projectiles immediately preceding the rocket motor body joint painting failure not inspected shall be inspected until thirty-two (32) consecutive projectiles have been tested without failure. Any projectile that fails for excessive paint on the bourrelet or on the rocket motor joint shall be classed defective and shall be removed from the lot.

4.4.3.6 Rocket selector cap. Projectiles received at this inspection station without a rocket selector cap shall not have a cap assembled, but shall be rejected and removed from the lot. Delay assemblies must be extracted and replaced.

4.4.3.7 Check test for possible deterioration of primer, percussion, M28 (see specification MIL-P-46565). If the total elapsed time between original acceptance of the M28 primer lot and the assembly of that lot into complete round cartridges exceeds two years, or if the M28 primers have been subjected to adverse conditions, however brief, at any time since previous tests, the applicable lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in specification MIL-P-46565 immediately before the M28 primer lot is assembled into the cartridges. The test shall be performed by the contractor on assemblies selected by the Government inspector at the facility assembling the complete cartridges (see 6.11).

4.4.4 Inspection equipment. Requirements for inspection equipment to perform examinations and tests are specified in the item detail specification and in specific contract clauses as applicable. See Section 6 of this specification for general provisions applicable to inspection equipment designs.

4.5 Methods of inspection.

4.5.1 Radoigraphic examination.

4.5.1.1 Determination of warhead defects. The warheads shall be subjected to radiographic examination. The radiographic beam shall be located in a plane which is perpendicular to the longitudinal axis of the warhead and tangent to the base of its internal cavity. The outline of the explosive charge shall be clearly defined at all points. Examination of the radiograph shall be made to determine

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compliance with applicable requirements. No images of lead masks, containers, fixtures, identifying marks or other extraneous features should project into the region of explosives under inspection, with the exception of specified penetrameters and radiation screening required to reduce film densities to readable levels in the thinner areas of the shell. Such screening should be free of any radiographically observable features in the projected area under inspection. In addition, the sensitivity of the radiographic technique shall be 2.0 percent and the minimum photographic density of the film used shall be no less than 1.25 at every point within the projected area of the explosive, including the base, with the maximum density no greater than to permit adequate inspection over the entire area of interest, as determined in accordance with 4.5.1.3 and 4.5.1.4. All defect-like features in the area of interest on a radiographic image shall be treated as defects unless identified as film or processing related or as acceptable metal parts irregularities. In the event the original radiograph leaves doubt as to a shell's acceptability because of difficulty in evaluating the size, location or classification of a defect, the warhead shall be rotated about its longitudinal axis from the original exposure, and additional radiographs taken for clarification. Acceptability in all cases shall be judged on the basis of the film giving the largest projection, except in the case where re-radiographing reveals multiple superimposed cavities each of which individually complies with the requirements. All additional radiographs of a shell for defect conditions will be identified on the film as retakes and stored; traceable to the original.

4.5.1.2 Determination of missing or improperly assembled motor body components. In addition to the warhead examination required by 4.5.1.1, the motor body section of the assembled projectile shall be subject to radiographic examination (see 4.4.3.3). In all exposures the radiographic beam shall form a 90 degree angle with the longitudinal axis of the projectile. Examination shall be made to determine compliance with the applicable requirements.

The aiming point of the beam shall be determined based on standard radiographs (see 4.5.1.5) which give most accurate results, and once the particular beam location is established, it shall not be changed unless new radiographic film standards are established and reviewed by the technical agency (see 4.5.1.5). Beam alignment will be established to optimize detection of gap at aft end of propellant grain.

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4.5.1.3 Check of sensitivity of radiographic technique. The sensitivity of radiographic technique shall be determined by means of penetrameters fabricated as specified herein. The penetrameters shall be placed on the radiographic source side of a warhead so that the 2T holes are at the points shown in Figure 2. Three penetrameters for each warhead shall be used. When radiographing a number of similar warheads the frequency of the sensitivity check shall be no less than one in each fifty radiographs, and on a rotating ring, the penetrameters shall be used at one station for each complete rotation of the ring.

The sensitivity of the radiographic technique shall be considered satisfactory when the image of the penetrameters, including the 2T holes, are perceptible in accordance with MIL-STD-453.

The penetrameters shall be made of steel having an approximate specific gravity of 7.8. The penetrameters shall be located as shown in Figure 2. For further details on construction of the penetrameter see MIL-STD-453.

4.5.1.4 Check of photographic density of film. The photographic density of the film shall be determined on the film in the region of minimum film density within the projected area of explosive charge. Film density shall be checked each hour of operation or fraction thereof, for each film processor. Greater frequency shall be required for any period during which difficulty is encountered in establishing the required densities. Photographic densities shall be measured with a diffuse density, photoelectric densitometer providing direct display of numerical density units. Accuracy of densitometer shall be maintained by calibrating with a photoelectric step tablet each hour immediately prior to measurements on the radiographic film. Each point of density measurement on the radiographs shall be identified by a circumscribed circle (made with a suitable pen or pencil) of no more than 1/4 inch in diameter. All photographic step tablets used in normal operation shall be checked for accuracy monthly by calibrating the densitometer with a reference photographic step tablet calibrated by National Institute of Standards and Technology (N.I.S.T.) or approved by ARDEC. Referenced tablet shall be kept available, but shall be protected from wear and damage and shall not be used in daily operations. Any photographic step tablet having severe surface damage or showing a deviation from previous calibrations by .05D or more shall be replaced with a new, unused step tablet.

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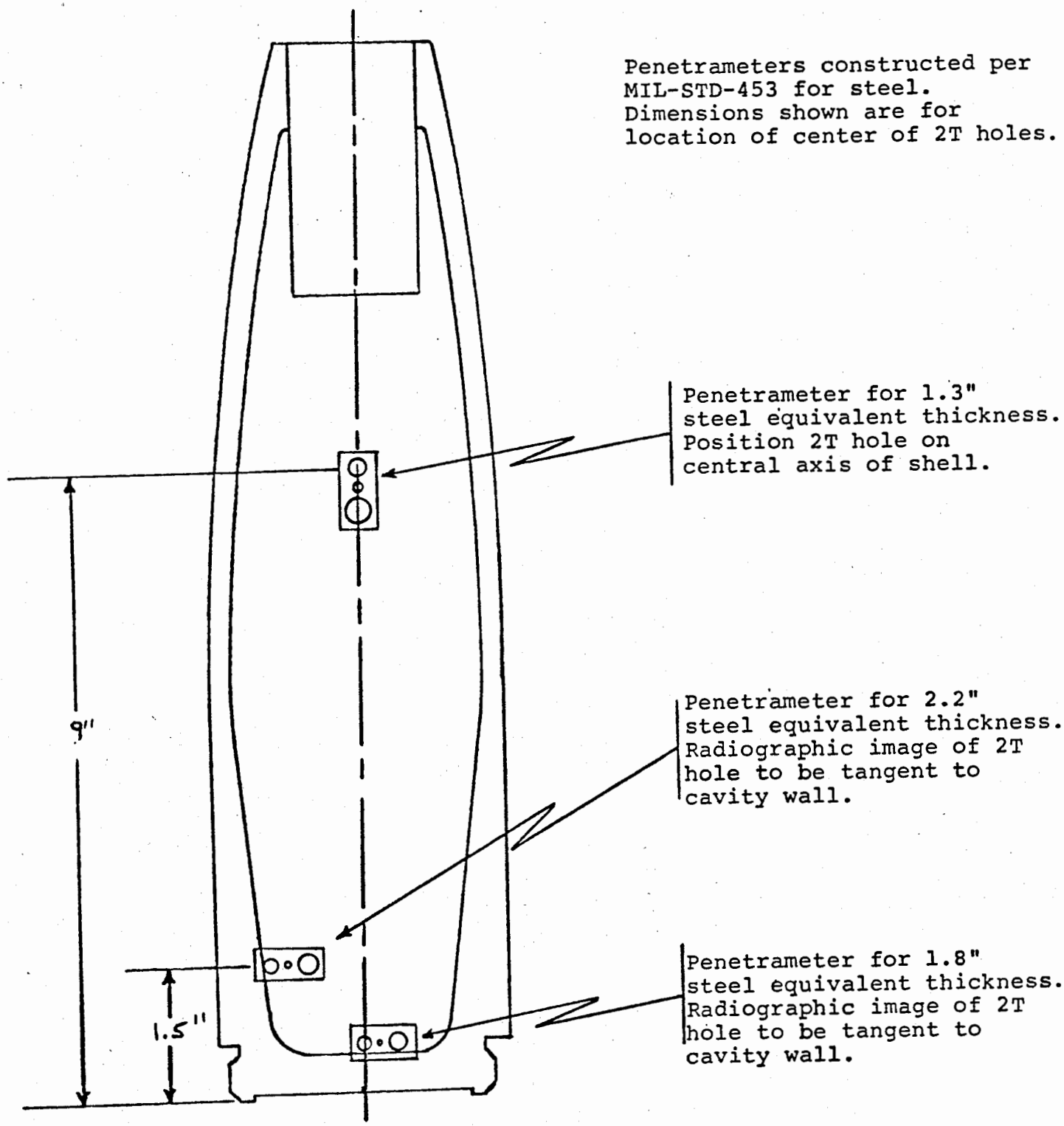


FIGURE 2. Proj. 105MM, HERA, XM913
Radiographic Penetrameter Requirements.

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4.5.1.5 Review of radiographic sensitivity and techniques. (See 6.9) The sensitivity of the radiographic techniques shall be reviewed by submitting one radiograph of the warhead standard and a copy of radiographic procedures prior to the start of initial production. In addition, a radiograph of the motor body section of an assembled projectile shall be randomly selected and submitted for review of the sensitivity of the inspection for proper assembly. These radiographs shall be taken and processed using the same techniques as will be used in production. In addition, one radiograph of each of the warhead standards and a radiograph of the motor body section shall be submitted for review once a month for the first three months of each contract and every three months thereafter.

4.5.1.6 Operator qualification. For operation of radiographic equipment, personnel shall be qualified and certified in accordance with MIL-STD-41Ø, Level I. For set-up and calibration of radiographic equipment, personnel shall be qualified and certified in accordance with MIL-STD-41Ø, Level II.

4.5.2 Torque tests. These tests shall be performed on the sample selected in accordance with 4.4.2.11 and 4.4.3.2. A thin line shall be drawn across the joint in question to aid in observing movement. The minimum assembly torque specified on the drawing for the rocket selector cap and the motor body shall be applied in the assembly direction. The minimum and maximum disassembly torques for the shipping plug shall be applied in the disassembly direction.

4.5.3 Determination of base separation. After heat treatment, the samples from each post cycle heat treatment lot shall be cooled to ambient temperature and sectioned in a manner to expose the interface between the cast and the interior base of the warhead at the appropriate center of the base. The warhead shall be positioned nose down or the cast shall be moved as far towards the nose with finger pressure as possible before measuring for base separation between the shell cavity and cast.

4.5.4 Ring gage. After touch-up or joint painting the sample projectiles per 4.4.3.6 shall be tested by passing through a ring gage manufactured to a maximum diameter. The ring gage is to pass over the bourrelet and rocket motor-warhead joint.

4.5.5 Cloth. At the time cloth is introduced to the sewing or cutting operation, an identification will be made for each roll to verify that proper material is used. Any cloth failing to be identified as proper material in accordance with applicable drawings and specifications shall be rejected from the lot.

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4.5.6 M30 propellant and potassium sulfate weights. The propellant charge, placed in a container, shall be weighed and then check weighed on a different balance. If the weighing is performed manually use another operator for check weighing. The weights shall be in accordance with the limits specified in 3.1.

4.5.6.1. Weight of contents of flash reducer.

4.5.6.1.1 Average weight of bags. A minimum of four (4) weighings must be made of empty bags during each shift to check for variation from nominal weight. Two (2) of these weighings shall be made within the first half of the shift and the other two weighings shall be made in the second half of the shift. Additional bag weights shall be taken, as required.

4.5.6.1.2 Procedure. Accurately weigh the assembled flash reducer on a balance. Then, calculate the net weight of the contents by difference using an average weight of the empty bags (as determined in 4.5.7.1.1). The calculation may also be performed by using the tare as marked on the individual bag instead of the average weight of the empty bags. The weight of the contents shall be within the total weight and combined tolerance of the propellant and potassium sulfate, as specified in 3.1.

4.5.7 Proving ground test. This test shall be performed at a Government proving ground in accordance with the applicable Acceptance Test procedure. The samples selected in accordance with 4.4.3.4 shall be fired in a M119 series cannon with muzzle break. The projectiles shall be fuzed with a M557 series PD fuze. Adequate photographic and instrumental coverage shall be provided to assure compliance with all possible requirements. If projectiles are received with "ballistic sample numbers' from the contractor then the projectiles shall be randomized with regard to test phase and order of firing. The following information pertaining to this test shall be obtained and reported for all phases as specified below.

a. Weapon and component data. All weapon and ammunition components used in support of this test shall be identified by model and lot or serial numbers. This information shall be reported on the firing record.

b. Observable attribute data shall be as follows:

1. Premature functioning.
2. Metal parts separation.
3. Rocket motor malfunction.
4. Impact functioning (information only).
5. Muzzle velocity

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c. Variable round-by-round data. Data specified in the Firing Record Information Sheets, see Figures 3 and 4 for sample forms, shall be obtained for each ballistic sample.

d. Meteorological (Met) data. Met data shall be obtained to an altitude of 14,000 meters at the start of testing (± 15 min) and at least every two hours throughout the test.

e. Radar data. Doppler radar (or a suitable alternative with Government approval) shall be obtained for each round fired until radar tracking cannot be maintained. Raw radar tapes shall be maintained a minimum of six months after release of the firing record.

f. Photographic coverage. Photographic coverage shall be adequate to assure compliance with the applicable requirements. Each projectile ballistically tested shall have four lines painted across the warhead/motor body joint (see 6.10). Photographic coverage shall be provided to determine relative movement between joints. Additionally, a second camera shall be positioned to ascertain ignition (burning) of the delay assembly columns.

4.5.7.1 Functioning/precision/muzzle velocities firings. All XM913 test and M760 master calibration cartridges shall be conditioned at 70°F ± 2 1/2° F using a XM229 series propelling charge conditioned at 70°F ± 2 1/2° F. All XM927 test and M760 master calibration cartridges shall be conditioned at 70°F ± 2 1/2° F, using a M67 series propelling charge at Zone 7 conditioned at 70° F ± 2 1/2° F. A 10 round sample shall be fired at a quadrant elevation of 450 mils. The 10 round group shall immediately be preceded by two (2) information control rounds. Time lapses of more than 15 minutes between consecutive rounds shall not be permitted and it is desirable that the 12 rounds (2 informational control samples plus 10 test samples) be completed within one (1) hour. Range probable errors shall be calculated for the 10 test round for determination of compliance with 3.6.1.2.1.

$$\text{Subgroup RPE (percent)} = \text{RPEj} \sqrt{\frac{\sum_{i=1}^{n} (X_i - X)^2}{n-1}} \times \frac{100 \times .6745}{\text{Ave. Range(j)}}$$

Where X = Individual horizontal ranges within each subgroup.

Where DF= Degrees of freedom n(1) + n (2) ...n(j) - number of subgroup.

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4.5.7.2 Safety. The XM913 projectiles for this test shall be divided into two equal groups. Each group shall be temperature conditioned for 16 hours minimum. Group 1 shall be conditioned and fired at
-50 degrees F at charge 8, XM229 series propelling charge conditioned at -50 degrees F. Group 2 shall be conditioned and fired at 145 degrees F at charge 8, XM229 series propelling charge conditioned at 145 degrees F. The XM927 projectiles for this test shall be divided into two equal groups. Group 1 shall be conditioned and fired at -50 degrees F at charge 7, M67 series propelling charge conditioned at -50 degrees F. Group 2 shall be conditioned and fired at 145 degrees F at charge 7, M67 series propelling charge conditioned at 145 degrees F. The two groups shall be fired at a convenient elevation selected by the proving ground.

5. PACKAGING

5.1 Unit and intermediate packaging.

5.1.1 Level A. Each cartridge shall be packaged in a PA 111 fiber container (dwg. 12624495) in accordance with dwg. 12624496.

5.2 Packing.

5.2.1 Level A. Each packaged fiber container (12624496) shall be packed in a PA 117 metal container in accordance with dwg. 12934900.

5.3 Marking.

5.3.1 Level A. (Fiber container). The fiber container shall be marked in accordance with dwg. 12624496.

5.3.2 Level A. (Metal container). The metal container outer pack shall be marked in accordance with dwg. 12934900 and MIL-STD-129.

5.4 Unitization.

5.4.1 level A. The packed metal container shall be unitized in accordance with dwg. 19048-4231/45-20PM1006.

5.5 Shipping. When ammunition from more than one lot is shipped at one time, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of the lots in transit. Ammunition data cards in accordance with MIL-STD-1167 shall be included.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The components and assemblies covered by this specification are intended for use on the cartridges, 105MM, HE, RA, XM913 and XM927. This specification contains requirements suitable for production.

6.2 Acquisition requirements. See MIL-A-48078.

6.3 Submission of inspection procedures or equipment designs for approval. See MIL-A-48078. Submit equipment designs as required to: Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAR-I (D), Picatinny Arsenal, New Jersey 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.4 Photographic densities. "American Standards for Diffuse Transmission Density" published by American Standards Association, 70 East 40th Street, New York City, New York, has been satisfactory as a source of information on film densities.

6.5 Clean explosive scrap. Clean explosive scrap is defined as scrap formed in kettles, risers and loading machines used in the pouring and cooling operations and scrap formed during the drilling operation.

6.6 Inspection lot size. It is noted that the size of inspection lots of components assemblies or item of delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of tests or deliveries, inspection lots of items of delivery may be equivalent to such contract quantities provided all of the lot formation criteria and sampling provisions of this specification are maintained.

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6.7 Proving ground test summary.

Test	Sample Size (see 4.4.3.4)		Requirement
Functioning	*30	10	a. Premature burst or metal parts separation (see 3.6.1.3). b. Individual range (see 3.6.1.2.2). c. Rocket Motor function (see 3.6.1.1). d. Range standard deviation (see 3.6.1.2.1).

* Applicable only for first three lots or tightened inspection.

Test	Sample Size (see 4.4.3.4)		Requirement
Safety (sample divided into 2 groups and fired hot and cold as per 4.3.2)	*30	14	a. Premature burst or metal parts separation (see 3.6.2). b. Individual range (see 3.6.2). c. Rocket Motor function (see 3.6.2).
TOTAL SAMPLE PROJECTILES REQUIRED	*60	24	

*Applicable only for first three lots or tightened inspection. Information-control projectiles are not included.

6.8 Submission of ballistic test data ammunition data cards. In addition to the normal distribution of records for loaded items provided by the Department of the Army, two copies of all ballistic test data and ammunition data cards shall be forwarded to: Commander, AMCCOM, ATTN: AMSMC-QAR-Q and SMCAR-FSA-IM, Picatinny Arsenal, New Jersey 07806-5000.

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6.9 Radiographic procedures. Radiographic procedures, standards and sample radiographs shall be submitted to: Commander, AMCCOM, ATTN: AMSMC-QAR-Q, Picatinny Arsenal, New Jersey 07806-5000 for review and approval.

6.9.1 Radiographic film sources. Approved list of film types and vendors for use in radiographing this projectile:

Eastman Kodak Company
1187 Ridge Road, West
Rochester, New York 14650

Film Types: AA, T, AX-1 or M

E. I. DuPont de Nemours and Company
Photo Products Department
Wilmington, Delaware 19898

Film Types: NDT-55, NDT-65 or NDT-70

GAF Corporation
140 West 51st Street
New York City, New York 10020

Film Types: 800 or 400

6.10 Ballistic test projectiles. All projectiles tested by a Government Proving Ground for Ballistic Acceptance Test shall have four (4) equally spaced lines painted, at the LAP manufacturing facility, across the Warhead/Motor Body joint. These lines shall be approximately one half inch wide by six inches long and positioned with three inches on the warhead and three inches on the motor body.

6.11 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research, Development, and Engineering Center (ARDEC) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.12 Rocket motor ignition failure at -50 degrees F. Degradation of rocket motor ignition reliability is expected at temperatures below -25 degrees F. When an ignition failure occurs at -50 degrees F during acceptance testing, the procuring agency must evaluate the following prior to determining lot disposition:

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a. The delay assembly shall have functioned properly as evidenced by review of radar data.

b. The failure shall not be related to a change in components or processes.

c. The rate of failure should not be significantly different from prior experience.

6.13 Notification of defects. The procuring activity will notified of any projectile rejected for metal parts defects found upon inspection of radiographs. The rejected projectile will be held pending disposition by the procuring activity.

6.14 Short range. Projectiles failing to achieve required range with proper motor function shall be considered critical defects. Projectiles failing to achieve required range due to improper motor ignition delay or motor burn shall be considered as major defects.

6.15 Metal parts separation. Metal parts separation which result in warhead impact short of 6000 meter safety zone for rocket-on fire, or which are not caused by rocket motor action, shall be considered critical defects. Metal parts separations caused by rocket motor action which result in warhead impact within the safety zone shall be considered major defects.

6.16 Cost of check test. The contracting officer shall arrange for the contractor to be reimbursed for the expense incurred in the performance of the check tests. The test shall be conducted at Government expense without cost to the contractor who loaded the primers or to the contractor assembling the cartridge, and shall not constitute a basis for rejection against either contractor, except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the primer lot was under the jurisdiction of either contractor.

6.17 Subject term (key word) listing.

Base separation
Delay parts
Loading of High Explosives
Metal packing containers
Projectile
Radiographic inspection
Surface cavitation
Torque, assembly

Custodian:
Army - AR

Preparing activity:
Army - AR

(Project 1320-AB94)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-70532 (AR)	29 October 1990

3. DOCUMENT TITLE
CARTRIDGES, 105MM, HE, RA, XM913 AND XM927 (WITH WARHEAD AND MOTOR BODY) LAP

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
US Army ARDEC Standardization & Specification Office	(1) Commercial (201) 724-6671	(2) AUTOVON 880-6671
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S Picatinny Arsenal, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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